

Library Fine Calculator

Automated system for overdue book tracking and fine evaluation

LIBRARY MANAGEMENT

OVERDUE TRACKING

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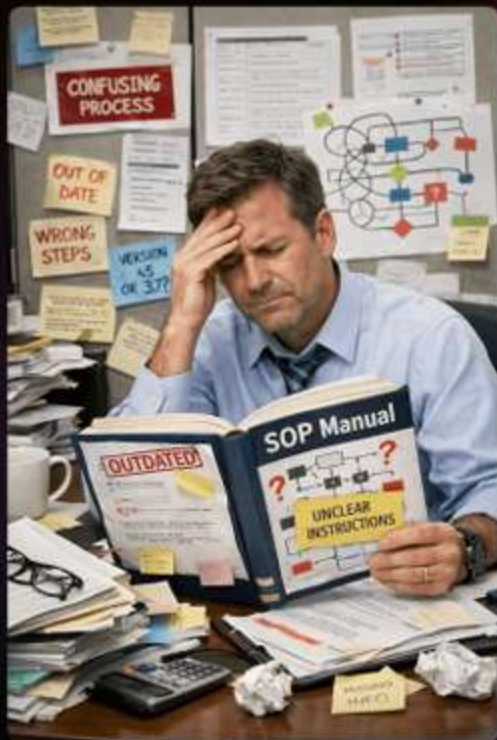
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Real-World Scenario: College Library



Problem

Manual fine calculation is error-prone and time-consuming — staff must check dates by hand, leading to mistakes and disputes.

Solution

An automated system that calculates fines instantly based on book return dates — no guesswork, no errors.

Impact

Reduces staff workload, improves accuracy, and enhances the overall user experience for students and librarians alike.



Project Objectives



User-Friendly Interface

Build a clean, intuitive web form that any student or librarian can use without technical knowledge.



Business Logic

Implement precise fine computation rules that mirror real library policies.



Seamless Integration

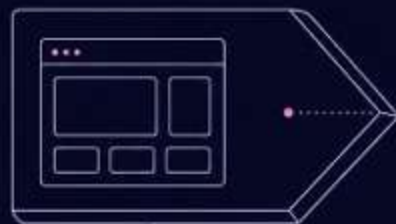
Connect frontend and backend smoothly so data flows without friction.



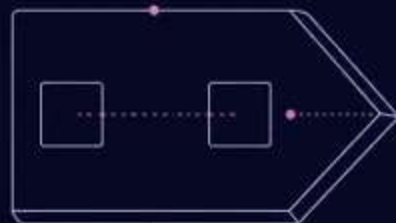
Edge Case Handling

Handle real-world exceptions like public holidays and grace periods gracefully.

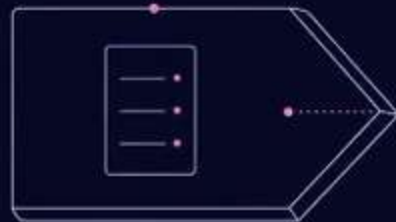
Technology Stack Overview



Frontend layer
with HTML, CSS,
JavaScript for interactive
web interface



Communication layer
with REST API or form
submission connecting
frontend to backend



Backend layer
with Java for business
logic and fine calculations

Frontend

HTML, CSS, JavaScript — builds the interactive web interface students and staff see and use every day.

Communication

REST API or Form Submission — the bridge that sends user input from the browser to the Java server securely.

Backend

Java — handles all business logic, fine computations, and data processing away from the browser.

Fine Calculation Logic

₹10

Per Day

Fine rate for the **first 7 days** overdue
— a fair, moderate penalty.

₹20

Per Day After

Escalated rate applied **after 7 days** to
encourage faster returns.

7

Day Threshold

The cutoff point where the fine **tier changes** from standard to escalated.



Exceptions: Weekends and public holidays can be excluded from overdue day count. A grace period may also be applied before fines begin.

System Architecture



This three-tier design separates concerns clearly — the browser handles presentation, Java handles logic, and the database (optional) handles persistence. Beginners can start with just the first two layers and add the database later.

Key Features to Implement

Real-Time Calculation

Fines are computed instantly as soon as the user submits the return date.

Fine Breakdown

Display a clear breakdown showing days overdue, rate applied, and total amount.

Multiple Books

Support batch calculation for students returning more than one overdue book at once.

History & Receipt

Generate a printable fine receipt and maintain a history log for the library's records.

Next Steps: Getting Started

Follow this roadmap to go from zero to a working Library Fine Calculator:

01

Set Up Project Folder Structure

Organise your files into `/frontend` and `/backend` folders from the start.

02

Create HTML Form with CSS Styling

Build the input form for book ID, issue date, and return date with clean styling.

03

Write JavaScript Validation & API Calls

Validate user inputs on the client side before sending data to the Java backend.

04

Build Java Backend with Fine Logic

Implement the tiered fine algorithm in Java and expose it via a servlet or Spring Boot endpoint.

05

Test with Sample College Library Data

Run real test cases using dates from your own college library to verify accuracy.



Thank You

We sincerely appreciate your time and attention during our presentation on the Library Fine Calculator.

Project Team:

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